Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Adaptive Vision-Language Models for Camera-Based Industrial Safety Monitoring and Safety-Aware Mobile Robot Task Planning in Dynamic Environments			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing (√), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Mechanical · Materials		Knowledge Services	
	Medium Category	Small Category	Medium Category	Small Category
	Robotics/ Automation Machinery	Robot Control and Intelligence Technologies	R&D / Engineering Services	Other R&D / Engineering-Related Technologies
Concept of Proposed Technology	The proposed technology is a two-step **adaptive Vision-Language Model (VLM)** framework designed for end-to-end industrial safety. Unlike conventional industrial safety systems that rely on static rules or predefined object categories, the proposed technology introduces a two-step adaptive VLM framework: • **Step 1 – Adaptive Safety Monitoring VLM**: Uses an adaptive VLM to automatically extract safety requirements and procedures for incidents from manuals, protocols, or natural language prompts, and determines safety violations via camera-based monitoring and provides corresponding action plans.			

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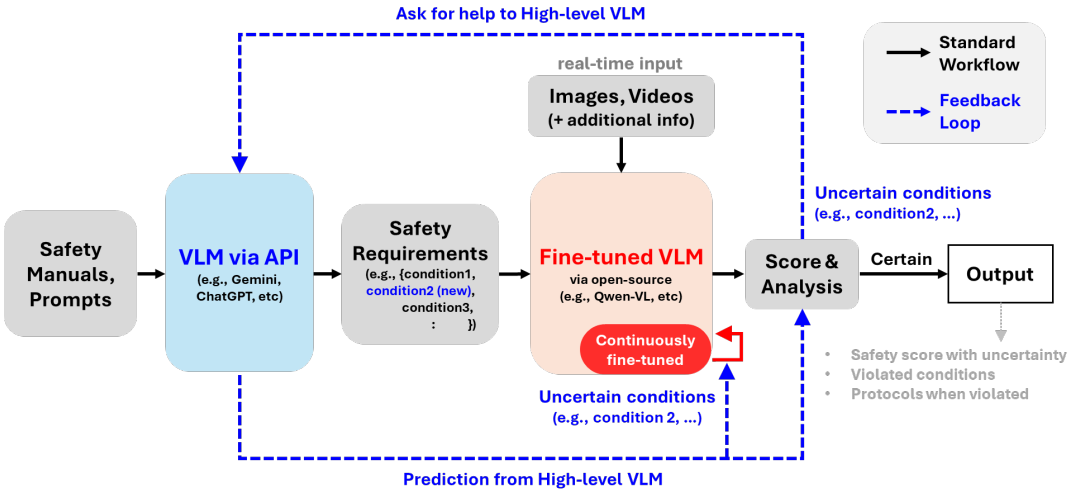
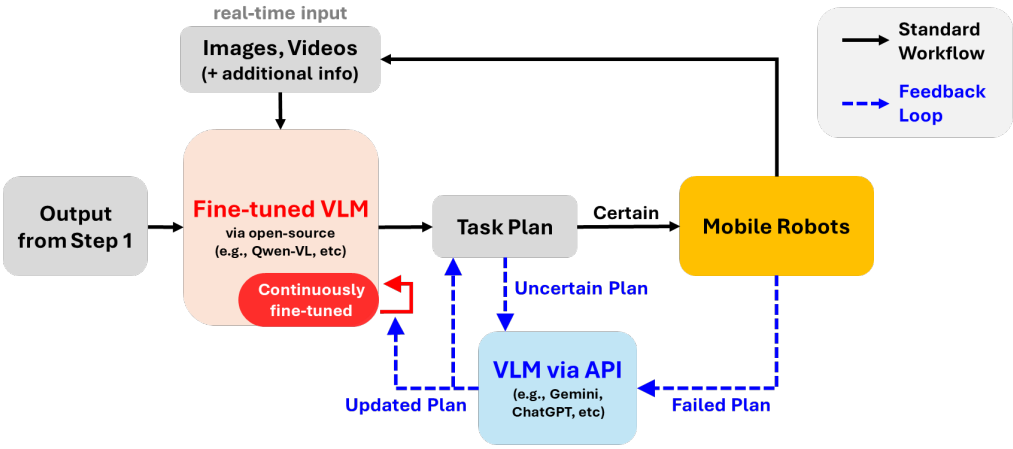


Step 1. Adaptive industrial safety monitoring VLM, camera-based)

- **Step 2 – Adaptive Safety-Aware Task Planning VLM**:
  Uses a separate adaptive VLM to generate and modify mobile robot task plans conditioned on safety-critical feedback (violations and procedural protocols) produced in Step 1.



Step 2. Adaptive safety-aware task planning VLM for mobile robots

Both models are designed to operate under **dynamic industrial environments**, where safety requirements, operational layouts, and task objectives continuously evolve. A unified continual learning and uncertainty-aware framework enables safe adaptation without full retraining, ensuring practical deployment in various settings.

Rationale for Joint R&D Project

Industrial safety is a safety-critical, domain-specific problem that cannot be effectively solved by generic AI models alone. Successful deployment requires tight coupling of advanced AI research capabilities with real industrial data, operational constraints, and validation environments.
This joint R&D project is necessary because:

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	• **Research institutes** contribute advanced expertise in:
	  ○ Vision-language models, large language models, and learning algorithms
	  ○ Uncertainty-aware prediction (Conformal Prediction)
	  ○ Task and motion planning algorithms
	• **Korean SME partners** provide:
	  ○ Real industrial safety requirements and operational protocols
	  ○ Camera data, robot task logs, and deployment environments
	  ○ Commercialization pathways and domain-specific validation
	
	Through international collaboration, the project accelerates the translation of **cutting-edge VLM research into deployable industrial products**, reducing development risk and shortening time-to-market for Korean SMEs.

Key Words	Vision-Language Models (VLMs), Industrial Safety Monitoring, Mobile Robot Task Planning, Continual Learning, Uncertainty-Aware Prediction (Conformal Prediction)

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (2)	End of the Project	Level (7)

R&D Objectives

1. Develop an adaptive VLM for industrial safety requirement understanding and camera-based safety violation detection, capable of interpreting dynamically changing safety rules defined in documents or user prompts.
2. Develop a separate adaptive VLM for safety-aware mobile robot task planning, which generates or revises robot task plans conditioned on safety-critical feedback.
3. Establish a unified continual learning and uncertainty-aware framework that enables both VLMs to adapt to uncertain decisions regarding the changes in environment, equipment, and safety requirements without full retraining.
4. Validate the integrated system in realistic industrial environments, achieving deployment-level reliability and readiness for commercialization.

R&D Target Specifications

Category	Target Specification
Safety requirement extraction accuracy	≥ 90% correctness for structured safety constraints from manuals/prompts
Safety violation detection (Step 1)	Precision ≥ 90%, Recall ≥ 95% under environmental changes
False alarm rate	≤ 10% in camera-based monitoring
Safety-aware task plan (Step 2)	≥ 95% successful task plan generation for standard scenarios
Safety constraint compliance	≥ 95% adherence to critical safety rules
Continual learning update	≤ 5% Forgetting rate on prior tasks during model adaptation; maintain performance over 10+ incremental task sequences
Reliability estimation	Calibrated confidence with statistically meaningful uncertainty bounds
Deployment readiness	Pilot-scale deployment with on-site validation

R&D Plan

Phase 1: Core Concept Validation and Baseline Development (Months 0–6)

- Formalize the two-step safety–planning **problem formulation** (Step 1 monitoring, Step 2 planning).
- Define structured representations for safety requirements extracted from documents and user prompts.
- Select and benchmark open-source **foundation VLMs** for industrial safety monitoring.
- Build initial datasets using industrial safety manuals, camera images, and video samples.
- Verify laboratory-scale feasibility of camera-based safety violation detection.

Phase 2: Adaptive Safety Monitoring VLM Development (Months 6–12)

- Develop the **adaptive VLM (Step 1)** for:
  - Safety requirement extraction from documents and natural language prompts.

- Camera-based safety violation detection under changing environments.
- Introduce **hierarchical supervision** using high-capacity VLMs to refine ambiguous labels.
- Implement **continual learning mechanisms** to update safety criteria without full retraining.
- Evaluate robustness against environmental changes (lighting, layout, equipment).

Phase 3: Safety-Aware Mobile Robot Task Planning VLM (Months 12–18)

- Design the **Step 2 adaptive VLM** specialized for mobile robot task planning conditioned on safety-critical feedback.
- Enable the model to:
  - Interpret structured safety feedback from Step 1.
  - Generate or revise task plans under safety constraints.
- Integrate safety-conditioned task planning with mobile robot simulators.
- Collect expert feedback for correcting unsafe or infeasible task plans.

Phase 4: Integrated System and Continual Learning Framework (Months 18–24)

- Integrate Step 1 (monitoring) and Step 2 (planning) into a unified system.
- Develop **uncertainty-aware mechanisms**, including:
  - Confidence calibration
  - Uncertainty estimation
  - Selective decision triggering under low confidence
- Implement continual co-adaptation of both VLMs during operation.
- Conduct pilot-scale testing in controlled industrial environments.

Phase 5: On-Site Validation and Performance Optimization (Months 24–30)

- Deploy the integrated system in **realistic industrial settings** with partner SMEs.
- Optimize system latency, robustness, and failure handling.
- Evaluate long-term adaptability to evolving safety requirements.
- Conduct reliability and stress testing under operational conditions.

Phase 6: Commercialization Readiness and Standardization (Months 30–36)

- Finalize deployment-ready system architecture.
- Support certification, standardization, and documentation for industrial adoption.
- Prepare commercialization strategies with Korean SME partners.
- Demonstrate scalability across multiple industrial sites and use cases.

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	The research institute will take primary responsibility for the **core AI technology research and algorithmic development** of the proposed system. Its roles include:
	
	• **Core algorithm and model development**
	  ○ Development of structured representations for safety constraints and procedures.
	  ○ Design and implementation of the adaptive Vision-Language Models (VLMs) for:
	  ▪ industrial safety requirement understanding and camera-based safety monitoring (Step 1)
	  ▪ safety-aware mobile robot task planning conditioned on safety-critical feedback (Step 2)
	
	• **Continual learning and uncertainty-aware framework**
	  ○ Develop continual learning approaches to enable continuous adaptation and improvement without full retraining.
	  ○ Development of uncertainty-aware prediction and confidence calibration mechanisms (e.g., conformal prediction).
	
	• **System architecture and integration logic**
	  ○ Design of the end-to-end system architecture linking safety monitoring outputs to task planning decisions.
	  ○ Definition of interfaces between perception, decision-making, and robot execution layers.
	
	• **Performance evaluation and validation methodology**
	  ○ Design of evaluation metrics and experimental protocols.
	  ○ Analysis of system reliability, robustness, and safety compliance.
	
	• **Technical guidance and knowledge transfer**
	  ○ Provide technical documentation, model specifications, and research findings to support technology transfer to SME partners.

Expected R&R of Korean SME Partner	The Korean SME partner will take primary responsibility for **industrial deployment, validation, and commercialization activities**, ensuring the developed technology meets real-world requirements. Its roles include:
	
	• **Provision of industrial data and environments**
	  ○ Supply real industrial safety manuals, operational protocols, camera data, and mobile robot task logs.
	  ○ Provide access to representative industrial environments for testing and validation.
	
	• **Deployment-oriented system development**
	  ○ Support integration of the developed AI models into existing industrial systems and workflows.
	  ○ Develop deployment infrastructure, including hardware integration and system interfaces.
	
	• **Testing, validation, and feedback**
	  ○ Conduct pilot-scale and on-site testing in operational environments.

	○ Provide expert feedback on safety violations, task planning outcomes, and system usability. • **Commercialization and standardization** ○ Lead certification, standard compliance, and productization efforts. ○ Develop commercialization strategies, including target markets and business models.

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	**Technological Outcomes**
	• A **commercialization-ready industrial safety AI system** integrating:
	  ○ camera-based safety monitoring, and
	  ○ safety-aware mobile robot task planning.
	• Two specialized **adaptive Vision-Language Models** with continual learning capability.
	• A **reliability- and uncertainty-aware framework** enabling trustworthy decision-making in safety-critical environments.
	• High-performance Vision-Language Model–based **safety reasoning** and **safety-conditioned task planning** solutions, adaptable to multiple industrial deployment
	
	Commercialization Impact
	• A deployable solution suitable for:
	  ○ manufacturing plants,
	  ○ construction sites,
	  ○ logistics facilities,
	  ○ smart factories, and
	  ○ other **safety-critical industrial environments**.
	• Reduced development risk and shortened time-to-market for Korean SMEs.
	• Cost-efficient and **economically feasible** deployment through the adoption of continually fine-tuned Vision-Language Models, reducing reliance on expensive large-scale VLM APIs.

6. References

Selected Papers from the Research Institute

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization